RR-J-800F
June 14, 1983
SUPERSEDING
Fed. Spec. RR-J-800E
October 5, 1973

FEDERAL SPECIFICATION

JUG, INSULATED; THERMAL EFFICIENT

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers thermal efficient jugs.

1.2 Classification.

1.2.1 Types, grades, and sizes. The jugs shall be of the following types, grades, and sizes, as specified (see 6.2).

Sizes are shown in gallons.

Type I Stainless steel inner shell

Grade A Stainless steel outer shell
Grade B Carbon steel outer shell

Size 1
Size 2
Size 3
Size 5

Type IV Plastic inner shell, material insulated

Grade B Carbon steel outer shell
Grade C Aluminum, plastic, or plastic and metal outer shell

Size 1
Size 2

FSC 7330

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-390	-	Plastic, Molding Material and Extrusion Material, Polyethylene (Low, Medium and High Density)
L-P-394	-	Plastic Molding Material, (Propylene Plastics, Injection and Extrusion)
L-P-398	-	Plastic Molding Material, Styrene-Butadiene
L-P-1183	-	Plastic Molding Material, Acrylonitrile-Butadiene-Styrene (ABS) Rigid
QQ-A-250/2	-	Aluminum Alloy 3003, Plate and Sheet
QQ-B-613	-	Brass, Leaded and Nonleaded; Flat Products (Plate, Bar, Sheet, and Strip)
QQ-B-626	-	Brass, Leaded and Nonleaded; Rod, Shapes, Forgings, and Flat Products With Finished Edges (Bar, and Strip)
QQ-W-423	-	Wire, Steel, Corrosion-Resisting
TT-E-485	-	Enamel, Semigloss, Rust-Inhibiting
TT-E-489	-	Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces)
TT-P-636	-	Primer Coating, Alkyd, Wood and Ferrous Metal
TT-P-1757	-	Primer Coating, Zinc Chromate, Low Moisture Sensitivity, Aluminum Surfaces
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic

Federal Standard:

FED-STD-123	-	Marking for Domestic Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

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(Single copies of this specification, other Federal specifications, standards, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-M-14	-	Molding Plastics and Molded Parts, Thermosetting
MIL-P-15328	-	Primer (Wash) Pretreatment, Blue (Formula No. 117 for Metals)
MIL-P-22748	-	Plastic, Polyethylene and Copolymers, High Density Molding and Extrusion Material

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

42 CFR 72	-	Interstate Quarantine
21 CFR 174	-	Food and Drug Administration

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

U.S. Department of Health and Human Services

List of Accepted Equipment for Interstate Carrier Use

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Sanitation Foundation (NSF)

Standard No. 2 - Food Service Equipment
Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

American Society for Testing and Materials (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
A 366 - Steel, Carbon, Cold-Rolled Sheet, Commercial Quality

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.2 Standard product. The jugs shall be the manufacturer's commercial product except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

3.3 Codes and standards. The jugs shall comply with the applicable requirements of either NSF Standard No. 2 or the Interstate Quarantine Regulations of the United States (42 CFR 72).

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3.3.1 Compliance. Prior to approval of the first article if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the jugs he proposes to furnish under this specification meet the applicable requirements of either NSF Standard No. 2 or the Interstate Quarantine Regulations of the United States as follows.

3.3.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished jug, or

(2) A certification for the jug issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical authority functioning in advisory capacity to the contracting agency of the specification, indicating that the jugs have been tested and conform to NSF Standard No. 2.

3.3.1.2 Interstate Quarantine Regulations of the United States. Acceptable evidence of meeting the requirements of the Interstate Quarantine Regulations of the United States shall be:

(1) A listing in the current edition or quarterly supplement of the "List of Accepted Equipment for Interstate Carrier Use", or

(2) A certificate or letter from the Public Health Service indicating that the jugs have been tested and found to conform to the applicable requirements of the Interstate Quarantine Regulations of the United States.

3.4 Materials and components. Materials (see 6.5), and components shall be as specified herein: Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial jugs provided the completed item complies with all provisions of this specification. All material must conform to the provisions of the Food and Drug Administration (21 CFR).

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3.4.1 Stainless steel.

3.4.1.1 Sheet and strip. Stainless steel sheet and strip shall conform to types 302, 304, or 305 of ASTM A 167.

3.4.1.2 Wire. Stainless steel wire shall conform to form I, composition 302 or 304, condition A of QQ-W-423.

3.4.2 Carbon steel. Carbon steel sheet shall conform to cold-rolled, commercial or drawing quality of ASTM A 366.

3.4.3 Aluminum alloy. Aluminum sheet shall conform to alloy 3003, temper H12 of QQ-A-250/2.

3.4.4 Brass. Brass for forming shall conform to copper alloy No. 260, quarter hard and brass for machining shall conform to composition 342 or 353 of QQ-B-613 or copper alloy No. 360 of QQ-B-626.

3.4.5 Plastic. Plastic shall be either virgin material or a mixture of virgin material and clean, unburned, reground plastic material from imperfect parts, sprues, runners or other scraps of the same composition as the virgin material.

3.4.5.1 Polyethylene, high density. High density polyethylene shall be linear type and shall conform to class A, grade 1 of MIL-P-22748.

3.4.5.2 Polyethylene, low density. Low density polyethylene shall conform to type I, class L, grade 2 or 3 of L-P-390.

3.4.5.3 Polypropylene. Polypropylene material shall conform to type III, grade A of L-P-394.

3.4.5.4 Polystyrene, modified. Modified polystyrene shall conform to type II, class 2 of L-P-398.

3.4.5.5 Acrylonitrile-butadiene styrene. Acrylonitrile-butadiene styrene shall conform to L-P-1183.

3.4.5.6 Phenolic heat-resistant molding plastic. Molding plastic shall conform to type MFH of MIL-M-14 except that reground scrap shall not be used and the electrical tests are not required.

3.4.6 Primer. Primer shall conform to TT-P-1757 or TT-P-636.

3.4.7 Enamel. Enamel shall conform to class B of TT-E-489 or type II of TT-E-485.

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3.5 Design and construction. The jugs shall be of the types, grades, and sizes as shown in 1.2.1 and shall have inner and outer shells joined together so as to form an insulating space between the shells. The jugs shall be insulated by vacuum or material or by a combination of vacuum or material in accordance with the manufacturer's standard commercial practices. The jugs shall have covers and the type I, size 2, 3, and 5 jugs shall have removable cover gaskets fabricated of rubber or plastic, or a combination of rubber and plastic. Handles or bails, and faucets shall be provided on each jug as specified herein. Faucet guards shall be provided on size 2, 3, and 5 jugs and are optional on the size 1 jug. The inner shell of all jugs and the outer shell of all type 1 jugs shall not leak when tested as specified in 4.4.2.2. The type I jugs shall not leak, break, increase in weight, or show evidence of parts separation when tested as specified in 4.4.5. All jugs shall meet the minimum capacity requirements specified in table I, when tested as specified in 4.4.5. The temperature loss of the water in all jugs shall not exceed the maximum allowable temperature loss specified in table I, when tested as specified in 4.4.5.

TABLE I. Capacity and temperature retention requirements

Sizes	Minimum capacity	Maximum allowable temperature loss
Size 1	1 gallon	50°F
Size 2	2 gallons	45°F
Size 3	3 gallons	40°F
Size 4	5 gallons	35°F

3.5.1 Type I jugs. The type I jugs shall have inner shells of stainless steel specified in 3.4.1.1 and shall be grade A or B, and size 1, 2, 3, or 5. Grade A jugs shall have an outer shell of stainless steel specified in 3.4.1.1. Grade B jugs shall have an outer shell of carbon steel specified in 3.4.2. The inner and outer shells shall be joined by welding or brazing (see 4.4.2.1). The bottom of the inner shell of the size 2, 3 and 5 jugs shall allow for drainage of the liquid. When specified for Air Force procurements (see 6.2), the type I, grade A, size 2 jugs shall have an overall height of 19 ± 1/4 inches and an overall diameter, exclusive of faucet, faucet guard, latches and cover, of 8-11/16 ± 1/8 inches and shall show no evidence of part separation or distortion when vacuum tested as specified in 4.4.5.

3.5.1.1 Covers, size 2, 3 and 5. Covers for type I, size 2, 3, and 5 jugs shall be of double wall construction of stainless steel specified in 3.4.1.1 with straight-through air vents and vent caps, The vent caps shall vent the

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jugs and cover the vent tubes when partially open. A device shall be provided to prevent loss of the vent cap. Covers shall be held in place over the jug opening by not less than two cam-locking devices and the covers shall be sealed by a removable gasket of rubber or plastic or a combination of rubber and plastic between the jug and the cover.

3.5.1.2 Cover, size 1. Cover for the type I jug shall be the same as specified in 3.5.1.1 or at the option of the manufacturer may be a threaded cover fabricated of plastic conforming to 3.4.5.3 or 3.4.5.4, stainless steel conforming to 3.4.1.1 or of nickel and chromium plated brass conforming to 3.4.4. Means shall be provided to equalize inner and outer jug pressures.

3.5.1.3 Stopper, type I, size 1. At the option of the supplier, and in addition to the cover required by 3.5.1.2, the type I, size 1 jug may have a vented, insulated, shell type stopper fabricated of plastic conforming to 3.4.5.2 or 3.4.5.4, or stainless steel conforming to 3.4.1.1.

3.5.1.4 Faucets. Faucets shall be made of plastic conforming to 3.4.5.3, 3.4.5.5, or 3.4.5.6, nylon with silicone seat, stainless steel conforming to 3.4.1.1, aluminum, or of nickel and chromium plated brass and shall be easily disassembled and assembled without the use of tools. The faucets for type I jugs shall be two position, self-closing and continuous flow type, except that at the option of the manufacturer the type I, grade B, size 2 jug may have a pushbutton faucet. The faucet nozzle opening and the inside diameter of the faucet tube of the size 1 and 2 jugs shall be not less than 5/16 inch and for the size 3 and 5 jugs shall be not less than 7/16 inch. The height between the bottom of the faucet and the bottom of the size 3 and 5 jugs shall be not less than 3-3/16 inches. Faucet guards shall be of the same material as the outer shell. Faucets on the size 1 jug without faucet guards (see 3.5) shall be constructed so as to prevent accidental opening.

3.5.1.5 Handles and bails. The type I, size 3 and 5 jugs shall have two handles of either a drop type, or a one piece rigid type mounted diametrically opposite each other on the breast or upper wall of the jugs. The handles shall be of the same material as the outer shell and shall be fastened to the jug by welding. The size 1 and 2 jugs shall have bails fabricated of plastic conforming to 3.4.5.3 or 3.4.5.4 or of stainless steel conforming to 3.4.1.1. The bail mounts shall be of the same material as the outer shell. At the option of the supplier, the handles specified for the size 3 and 5 jugs may be substituted for the bails on size 1 and 2 jugs. The handles and handle mounts of size 3 and 5 jugs shall not break, pull loose or become permanently deformed when tested as specified in 4.4.5. The bails and bail mounts or handles on the size 1 and 2 jugs shall not break, pull loose or become permanently deformed when tested as specified in 4.4.5.

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3.5.2 Type IV jugs. The type IV jugs shall have inner shells of plastic conforming to 3.4.5.1 or 3.4.5.3 and shall be grade B, or C, and size 1 or 2. The grade B jug shall have an outer shell of carbon steel conforming to 3.4.2. The grade C jugs shall have outer shells of aluminum alloy conforming to 3.4.3, or of plastic conforming to 3.4.5.1 or 3.4.5.3, or of plastic and carbon steel, or of plastic and brass. Brass, if used, shall conform to 3.4.4. The inner and outer shells shall be joined in accordance with the manufacturer's standard commercial practice. The type IV jugs shall have bails, stoppers or covers, and faucets as specified in 3.5.2.1 through 3.5.2.6. The bails and bail mounts shall not pull loose, break, or become permanently deformed when tested as specified in 4.4.5.

3.5.2.1 Details of components. Details of components specified hereinafter apply to the type IV jugs. Components shall be attached to the jugs in accordance with the manufacturer's standard commercial practice.

3.5.2.2 Bails. The bails shall be of any of the following:

(a) Aluminum alloy conforming to 3.4.3.
(b) Plastic conforming to 3.4.5.3, 3.4.5.4, or of vinyl.
(c) Steel strip; steel wire with wood or plastic handle; or steel tube.
(d) Stainless steel sheet or strip conforming to 3.4.1.1 or stainless steel wire conforming to 3.4.1.2.

3.5.2.3 Stoppers. Stoppers when used shall be of plastic conforming to 3.4.5.2, 3.4.5.3, or 3.4.5.4 or of stainless steel conforming to 3.4.1.1 and shall be an insulated shell type.

3.5.2.4 Covers. Covers shall be of plastic conforming to 3.4.5.2, 3.4.5.3, or 3.4.5.4; stainless steel conforming to 3.4.1.1; or of nickel and chromium plated brass.

3.5.2.5 Spout covers for shoulder spout jugs. Spout covers for jugs having shoulder mounted spouts shall be of plastic conforming to 3.4.5.3 or 3.4.5.4.

3.5.2.6 Faucets. Faucets shall be of plastic conforming to 3.4.5.3 or 3.4.5.5; nylon with silicone seat; stainless steel conforming to 3.4.1.1; or nickel and chromium plated brass.

3.6 Finish. The carbon steel outer shells of the type I, grade B jugs and the type IV, grade B jugs and the carbon steel portions of the outer shells of the type IV, grade C jugs shall be pretreated in accordance with the requirements of MIL-P-15328 and coated with primer conforming to 3.4.6, then coated with enamel conforming to 3.4.7.

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3.7 Marking.

3.7.1 Identification. Identification marking shall be in accordance with the manufacturer's standard commercial product.

3.7.2 Capacity. The capacity of the jugs, in gallons, shall be permanently and legibly marked on the side or bottom of each jug.

3.7.3 Caution. Each jug shall be permanently, conspicuously, and legibly marked with the following:

"DO NOT USE FOR MILK OR MILK PRODUCTS"

3.8 Workmanship. All components and assemblies of the jugs shall be free from dirt and other harmful extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts and stampings, if used, shall be free of sand, fins, pits, blowholes and sprues. External surfaces shall be free of sharp edges and corners.

3.8.1 Metal fabrication. Metal used in the fabrication of the jugs shall provide for original quality surface finish and shall be free from kinks and unspecified bends. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major nature shall be made with dies or fixtures to insure uniformity of size and shape.

3.8.2 Welding and brazing. The surfaces of all parts to be welded or brazed shall be free from rust, scale, paint, grease and other foreign matter. Welds and brazes shall be smooth and free of cracks, burn holes, undercuts and incomplete fusion. All scale and flux shall be removed from the finished weld and braze areas.

3.8.3 Fastening device. Rivet holes shall be accurately punched or drilled and shall have burrs removed. Rivets shall fill the holes completely and the heads and peened ends shall be in full contact with the surfaces of the joined members.

3.8.4 Enamel finish. The enamel finish shall be dry, complete, uniform in coverage and color and free of imbedded foreign matter, smudges, finger marks, smear, orange peel, drips, sags, wrinkles, blisters and runs. There shall be no areas of poor or no coverage or any visible evidence of rust showing through the finish coat.

3.8.5 Plastics. Molded plastic components shall be clean, uniform in color, and free of rough edges, scratches, fractures, chips, dents, holes, and other deformities.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.4.5. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling or inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 In-process examination. All joined inner and outer shells of the type I jugs shall be examined to determine compliance with the welding and brazing requirements of 3.5.1 and 3.8.2. Whenever nonconformance is noted, correction shall be made to the items affected, the lot in progress, and to the operation. Parts which cannot be corrected shall be removed from production.

4.4.2.2 In-process testing. All inner and outer shells of the type I jugs and all inner shells of the type IV jugs shall be tested for leakage as specified in 4.6.1 to determine compliance with the non-leaking requirements of 3.5. Any nonconformance with specified requirements shall be cause for rejection of the affected item.

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4.4.3 End item visual examination. The end item shall be examined for the defects listed in table II. The lot size shall be expressed in units of jugs. The sample unit shall be one jug. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. End item defects

Examine	Defect	Classification Major	Minor
Standard product	Item is not in accordance with manufacturer's standard product except for changes necessary to comply with specification requirements	X	
Finish:			
Stainless steel shells	Not as specified	X	
Carbon steel shells	Not enameled finish	X	
Enameling	Runs, wrinkles, foreign material imbedded, orange peel, areas of no coverage, scratches, not dry, not uniform		X
Construction and workmanship (all components)	Any part or component not fabricated of the applicable referenced materials	X	
	Metal corners not rounded; burrs, slivers, or sharp edges	X	
	Dents or deformities 1/2 inch or less in length		X
	Dents or deformities greater than 1/2 inch long	X	
	Fracture	X	
	Any component missing	X	
Welding	Burn holes, cracks, fractures, incomplete fusion, not smooth	X	
	Scale or flux deposits		X
Drainage	Bottoms of the inner shell of size 2, 3 and 5 jugs do not allow drainage	X	

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TABLE II. End item defects (cont'd)

Examine	Defect	Classification Major	Minor
Covers:			
Sizes 2, 3, and 5	Not double wall construction	X	
	Not vented	X	
	Gaskets not removable (type I)	X	
	Less than two cam-locking devices (type I)	X	
Type I, size 1	Means not provided to equalize inner and outer jug pressures	X	
Stopper	Not vented (when used)		X
Handles (as applicable)	Not type or location specified	X	
Bails (as applicable)	Not type specified	X	
ıcets	Not type specified	X	
Marking	Not as specified		X

4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expessed in units of jugs. The sample unit shall be one jug. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The end item shall be tested for the characteristics shown in table III in the test sequence indicated. The lot size shall be expressed in units of jugs. The sample unit shall be one jug. The inspection level shall be S-2 for all tests except jug durability and impact resistance. For jug durability and impact resistance, one jug out of each lot of 500 or fraction thereof shall be tested. Any test failure shall be cause for rejection of the lot.

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TABLE III. End item tests

Characteristic	Test sequence	Requirement paragraph	Test method paragraph
Capacity	1	3.5	4.6.6
Bail handles and mounts strength (type I, sizes 1 and 2)	2	3.5.1.5	4.6.7
Bails and mounts strength (type IV)	2	3.5.2	4.6.7
Handle and mounts durability (type I, sizes 3 and 5)	3	3.5.1.5	4.6.3
Jug durability (type I)	4	3.5	4.6.3
Impact resistance (type I)	5	3.5	4.6.4
Vacuum exposure (type I, grade A, size 2 for Air Force only)	6	3.5.1	4.6.9
Submersion resistance (type I)	7	3.5	4.6.5
Jug leakage (type I)	8	3.5	4.6.2
Heat retention	9	3.5	4.6.8

4.4.6 Packaging inspection. An examination shall be made to determine that preservation, packing and marking requirements comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL ,expressed in terms of defects per hundred units shall be 2.5.

Examine	Defect
Markings (exterior and interior)	Missing; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged or not as specified
Contents	Number per carton is not as specified

4.4.7 Palletization examination. An examination shall be made to determine that palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application

4.5 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.6 Methods of inspection.

4.6.1 Shell leakage test. Seal the inner shell of all jugs and the outer shells of all type I jugs and apply an air pressure of not less than 15 pounds per square inch gage for not less than 1 minute inside of the shell. Observe the air pressure gage to determine conformance with the nonleakage requirements of 3.5. Any nonconformance with specified requirements shall be cause for rejection of the shell.

4.6.2 Jug leakage test. The jug with cover in place and vent capped or plugged and at ambient temperature, shall be submerged in water having a temperature of not less than 150°F for not less than 2 minutes. The water shall be observed for air bubbles caused by a leak from inside of the jug and the jug shall be removed from the water and examined to determine conformance with the air tightness requirements of 3.5. Inability of the jug to remain submerged without emitting air bubbles shall constitute failure of the test.

4.6.3 Durability test. The jug shall be filled with water and the stopper or cover replaced. The jug shall then be placed in a device that will lift it by the handle(s) to a height of not less than 3 inches and drop it onto a steel base at a rate of not less than 1,500 drops per hour for not less than 1 hour. The jug shall then be examined for breaks and part separations to determine conformance with the strength requirements of 3.5 and 3.5.1.5. Inability of the jugs to undergo the test without breaking, parts separation, or permanent deformation shall constitute failure of the test.

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4.6.4 Impact resistance test. The weight of the empty jug shall be determined and recorded. The jug shall then be filled with water and the stopper or cover replaced. The water filled jug shall then be dropped five times onto a steel base. The jug shall be in an upright and level position for the first drop. A 3-inch cube of wood shall be placed under the bottom edge of the jug to vary the angle of the jug for the remaining four drops. The block shall be moved 90 degrees clockwise after each drop. The clear height of drop shall be not less than 37 inches. No guiding or restraining devices shall be attached to or be in contact with the jug during the drop. After the fifth drop the jug shall be emptied and weighed. The weight before dropping shall be compared with the weight after dropping to determine conformance with the strength requirements of 3.5. Any weight increase, breaking or parts separation shall constitute failure of the test.

4.6.5 Submersion resistance test. The weight of the empty jug shall be determined and recorded. The stopper or cover shall then be removed and the jug completely submerged under not less than 3-1/2 inches of water for a period of not less than 24 hours. The jug shall then be removed from the water and emptied and dried. The stopper or cover shall then be replaced and the jug shall be reweighed. The weight before testing shall be compared with the weight after testing to determine compliance with weight requirements of 3.5. Any weight increase shall constitute failure of the test.

4.6.6 Capacity test. The jug shall be filled with water to full capacity and the cover or stopper replaced. The water shall then be drained or poured into a standard liquid measure. The measured capacity shall then be compared to the minimum capacity shown in table I to determine conformance with the minimum capacity requirements of 3.5. Inability of the jug to meet the capacity requirements of table I shall constitute failure of the test.

4.6.7 Bails, handles, and mounts strength test. The size 1 and 2 jugs equipped with either bails or handles shall be secured in an inverted position and a test weight hung from the center of the bail and then from each of the bail mounts or from each of the handles as applicable for not less than 1 minute. The test weight for the size 1 jug shall weigh not less than 40 pounds and the test weight for the size 2 jug shall weigh not less than 60 pounds. The bail, bail mounts and handles shall then be examined to determine conformance with the strength requirements of 3.5.1.5 or 3.5.2, as applicable. Any permanent deformation or breaking of bails, mounts, handles, or handle mounts shall constitute failure of the test.

4.6.8 Heat retention test. The jug shall be filled with water having a temperature from 5° to 9°F below the boiling point and let stand for 5 minutes. The jug shall then be emptied and immediately refilled with water of the same temperature, the stopper or cover replaced, and the jug allowed to stand in a temperature of 72°F plus or minus 2°F for not less than 6 hours. The temperature loss of the water shall be determined and shall show no greater loss than the maximum temperature loss specified in table I. Temperature loss exceeding that specified shall constitute failure of the test.

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4.6.9 Vacuum exposure test. The type I, grade A, size 2 jugs intended for Air Force use (see 3.5.1) with covers removed, and empty, shall be placed in a vacuum chamber and a vacuum of not less than 3-1/2 inches of mercury shall be applied for a period of not less than 3-1/2 hours. The jug shall then be allowed to stabilize at atmospheric pressure for not less than 1-hour. The process shall be repeated six times and the jug shall then be examined for parts failure, parts separation and distortion to determine compliance with the requirements of 3.5.1. Any nonconformance with specified requirements shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Removable cover gaskets shall be placed within the interior of the size 2, 3, and 5 gallon jugs. The cover shall be secured in place on the jug. Each jug shall be unit packed in a fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Each jug shall be cushioned and completely wrapped in cellulose cushioning material conforming to type II, class B of PPP-C-843. Scored and folded sheets of the same material as the box shall be used to protect the faucet within the box. Closure shall be in accordance with the appendix of PPP-B-636.

1.2 Commercial. Each jug shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Jugs of one type, grade, and size only, preserved as specified in 5.1, shall be packed in the quantities as specified in table IV in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636.

TABLE IV. Quantity per shipping container

Size of jug	Quantity of jugs per shipping container
1	6
2	4
3	2
5	2

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5.2.2 Level B packing. Jugs of one type, grade, and size only, preserved as specified in 5.1, shall be packed in the quantities as specified in table IV in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. Jugs preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Palletization. When specified or for rail carload lots (see 6.2), jugs of one type, grade, and size, packed as specified shall be palletized in accordance with load type I of MIL-STD-147. The pallet shall conform to figure 2 of the standard or shall conform to PPP-P-1660, type to carry the load, style 5, class c, grade 2, size as applicable in table II. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with FED-STD-123 or MIL-STD-1188, as applicable. Special handling marking requirements shall apply.

5.4.2 Military agencies. In addition to any special marking required by the contract or order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable. Special handling marking requirements applicable to fragile items, arrows, and the word "UP" shall apply.

5.4.3 Additional marking. Each size 2, 3, and 5 gallon jug with removed gasket shall have a tag containing the legend "INSTALL GASKET BEFORE USING" secured to the exterior of the jug.

6. NOTES

6.1 Intended use. The insulated jugs are intended for use in hospitals, mess halls, aircraft, and other locations and for use in the field for keeping liquids hot or cold.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, grade, and size of jug required (see 1.2.1).
(c) When a first article is required (see 3.1, 4.3 and 6.4).
(d) When the type I, grade A, size 2 jugs are for Air Force procurement (see 3.5.1).
(e) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(f) When weather-resistant fiberboard is required for level B pack (see 5.2.2.1).
(g) When palletization is required (see 5.3).

6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one complete jug. The contracting officer should include specific instructions in all procurement documents regarding arrangements for inspection and approval of the first article.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.4).

MILITARY INTERESTS:

Custodians

Army - GL
Navy - SA
Air Force - 99

Review Activities

Army - MD
Navy - MS
Air Force - 84

User Activities

Navy - CG, MC, YD

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
VA - OSS

PREPARING ACTIVITY:

Army - GL

Project No. 7330-0555

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